depending on what node (or nodes) it happens to be running on. To solve this problem, Kubernetes has the concept of a *service*, which provides a way for pods to find other pods within the cluster. Each service running within Kubernetes is given a unique virtual IP address that is unique to that service and Kubernetes keeps track of which pods implement that service. In a microservice architecture you would register each microservice as a separate Kubernetes service. A process running in a container can call another microservice's API by making a network request to the virtual IP address corresponding to that service. Kubernetes will intercept the request and redirect it to a pod that implements the service.

> DEFINITION A Kubernetes ***service*** provides a fixed virtual IP address that can be used to send API requests to microservices within the cluster. Kubernetes will route the request to a pod that implements the service.

As pods and nodes are created and deleted, Kubernetes updates the service metadata to ensure that requests are always sent to an available pod for that service. A DNS service is also typically running within a Kubernetes cluster to convert symbolic names for services, such as `payments.myapp.svc.example.com`, into its virtual IP address such as `192.168.0.12`. This allows your microservices to make HTTP requests to hard-coded URIs and rely on Kubernetes to route the request to an appropriate pod. By default, services are accessible internally only within the Kubernetes network, but you can also publish a service to a public IP address